INCH-POUND
MIL-P-24441/19A(SH)
23 July 1991
SUPERSEDING
MIL-P-24441/19(SH)
31 August 1987

MILITARY SPECIFICATION SHEET

PAINT, EPOXY-POLYAMIDE, ZINC PRIMER, FORMULA 159, TYPE II

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-P-24441.

FORMULA. This formula covers zinc epoxy-polyamide paint designated Navy Formula 159, type II, for interior or exterior use. The paint shall consist of the ingredients specified in the quantities specified.

Component A

	Pounds
Polyamide 2/	8
Polyamide adduct 3/	106
Naphtha 4/	30
Paint thinner 5/	13.2

Component B

	Pounds
Zinc pigment 6/	1900
Epoxy resin 7/	190
Thixotrope 1/	15
Naphtha 4/	123
Paint thinner 5/	58.7

See footnotes on next page.

AMSC N/A
FSC 8010
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Formula 159 thinner

	Pounds
Naphtha 4/	622
Paint thinner 5/	198

QUANTITATIVE REQUIREMENTS. The paint shall meet the following quantitative requirements and the qualitative requirements of section 3 of the general specification. Components A and B shall be mixed 1:4 by volume for mixed components tests. Tests shall be performed in accordance with the general specification.

Requirements 8/	Component A		Component B		Mixed components	
	Min	Max	Min	Max	Min	Max
Pigment content, percent	---	---	81.5	85.5	---	---
Volatiles, percent	42.8	44.3	8.0	8.4	---	---
Nonvolatile vehicle, percent	53.7	57.7	8.3	8.7	---	---
Weight per gallon, pounds (lb/gal)	7.3	8.4	27.5	28.4	---	---
Water, percent	---	0.5	---	0.2	---	---
Flash point, °F	99	---	110	---	99	---
Sag resistance, mils	---	---	---	---	12	---
Dry-hard, hours at 73°F, 3 mils wet film	---	---	---	---	---	8
Set-to-touch time, hours at 73°F, 3 mils wet film	---	---	---	---	---	2
Pot life, hours at 73°F	---	---	---	---	4	---
Consistency, grams	---	---	250	500	150	300
VOC, ASTM D 3960, lb/gal	---	---	---	---	---	2.5

1/ The amount and type of thixotrope shall be selected by the manufacturer to meet all requirements of the general specification and this specification sheet.
2/ GENAMID 2000, Henkel Corporation; UNIREZ 2810, Union Camp; AZAMIDE 600, AZ Products; ANCAMINE 507, Pacific Anchor Chemical Corp.; EPOTUF SF7791, Reichhold Chemical; TriChem 9200, Trimont Chemicals.
3/ VERSAMID 280B75, Henkel Corporation; UNIREZ 2180B75, Union Camp; AZAMIDE 680B75, AZ Products; ANCAMINE 700B75, Pacific Anchor Chemical Corp.; EPOTUF SF7792, Reichhold Chemical; TriChem 9280-B-75, Trimont Chemicals.
4/ Conforming to ASTM D 3734, type I. In the development of this product, Amsco Super High Flash Naphtha was used.
5/ Conforming to TT-T-291, type II, grade A.
6/ Zinc No. 555, American Smelting & Refining; Zinc No. 422, New Jersey Zinc; or Intermediate No. 32 non-gassing, Pacific Smelting Company.

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7/ EPON 828, Shell Chemical Co.; ARALDITE 6010, Ciba-Geigy; DER 331, Dow Chemical Co.; AZEPOXY 128, AZ Products; EPO-TUF 37-140, Reichhold Chemical; EPI-REZ 510, Cleanese; TriChem 728, Trimont Chemicals.

8/ Tests on mixed component shall consist of one volume Component A mixed with four volumes Component B then reduced with 0.3 lb/gal thinner by volume. Thinner used shall be formula 159 thinner.

Use of alternate ingredients in this formula must have prior approval by the Naval Sea Systems Command. Approval will be based on review of data showing equivalent physical and chemical characteristics to the specified ingredient. It will be necessary to demonstrate that paint made using the ingredient will conform to all requirements of this specification.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 8010-N238)

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